ARJUNA JEE 2027

Lecture -10 | Mathematics

TRIGONOMETRIC FUNCTIONS

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Topics *to be covered*

A **Cosine Series**

B **Conditional Identities**

C **Question Practice**

D





LAST CLASS RECAP

Answer the following Questions:

(1) $\sin 18° = \frac{\sqrt{5}-1}{4}$

(2) $\cos 36° = \frac{\sqrt{5}+1}{4}$ ✓

(3) **How to find value of sin 36° & sin 72°**

$\sin 36° = \sqrt{1-\cos^2 36°}$ ✓

$\sin 72° = \cos 18° = \sqrt{1-\sin^2 18°}$ ✓

(4) $\tan\left(\frac{\pi}{8}\right) = \sqrt{2}-1$.

LAST CLASS RECAP

Answer the following Questions:

(i) $\sin(A+B)\cdot\sin(A-B) = \sin^2A - \sin^2B$

(ii) $\cos(A+B)\cdot\cos(A-B) = \cos^2A - \sin^2B$

(iii) $\sin\theta\cdot\sin(60^\circ+\theta)\cdot\sin(60^\circ-\theta) = \frac{1}{4}\sin3\theta$

(iv) $\cos\theta\cdot\cos(60^\circ+\theta)\cdot\cos(60^\circ-\theta) = \frac{1}{4}\cos3\theta$

(v) $\tan\theta\cdot\tan(60^\circ+\theta)\cdot\tan(60^\circ-\theta) = \tan3\theta$

H.W. Discussion

Question [JEE Mains-2023]

The value of $36(4\cos^2 9^\circ - 1)(4\cos^2 27^\circ - 1)(4\cos^2 81^\circ - 1)(4\cos^2 243^\circ - 1)$ is

A 18

B 36

C 54

D 27

$$36 \times \frac{\sin 27^\circ}{\sin 9^\circ} \times \frac{\sin 81^\circ}{\sin 27^\circ} \times \frac{\sin 243^\circ}{\sin 81^\circ} \times \frac{\sin(9 \times 243^\circ)}{\sin 243^\circ}$$

36×1.

$729^\circ = 720^\circ + 9^\circ$

$= 4 \times 180^\circ + 9^\circ$

$= 4\pi + \theta$.

$\sin(4\pi + \theta) = \sin\theta$.

$4\cos^2\theta - 1$

$= 4(1 - \sin^2\theta) - 1$.

$= (3 - 4\sin^2\theta) \cdot \frac{\sin\theta}{\sin\theta}$.

$= \frac{\sin 3\theta}{\sin\theta}$.

If $\sin^2(10^\circ)\sin(20^\circ)\sin(40^\circ)\sin(50^\circ)\sin(70^\circ) = \alpha - \frac{1}{16}\sin(10^\circ)$, then $16 + \alpha^{-1}$ is equal to

$$\left(\sin 10^\circ\right)\left(\sin 10^\circ \sin 50^\circ \sin 70^\circ\right) \times \sin 20^\circ \sin 40^\circ$$

$$\left(\sin 10^\circ\right)\ \frac{1}{4}\sin(3\times 10^\circ) \times \frac{1}{2}\left(2\sin 20^\circ \sin 40^\circ\right)$$

$$\left(\sin 10^\circ\right)\times \frac{1}{4} \times \frac{1}{2} \times \frac{1}{2}\left(\cos(20^\circ) - \frac{1}{2}\right) = \frac{1}{16}\sin 10^\circ \cos 20^\circ - \frac{1}{32}\sin 10^\circ$$

$$\frac{1}{16}\left(\frac{1}{4} - \frac{1}{2}\sin 10^\circ\right) - \frac{1}{32}\sin 10^\circ$$

$$\underbrace{\frac{1}{64}}_{\alpha} - \frac{1}{16}\sin 10^\circ$$

$$16 + \frac{1}{\alpha} = 16 + 64 = 80$$

$$\frac{1}{2}\left(2\sin\overbrace{10^\circ}^{A}\cos\overbrace{20^\circ}^{B}\right) = \frac{1}{2}\left(\sin(10^\circ+20^\circ) + \sin(10^\circ-20^\circ)\right)$$

$$= \frac{1}{2}\left(\frac{1}{2} - \sin 10^\circ\right)$$

$$= \boxed{\frac{1}{4} - \frac{1}{2}\sin 10^\circ}$$

If $\alpha = \cot 70^\circ + 4\cos 70^\circ$, $\beta = \sin^2 24^\circ - \cos^2 84^\circ$,

$\gamma = \cos^2 73^\circ + \cos^2 47^\circ + \cos 73^\circ \cos 47^\circ$,

$$\delta = \sin\frac{\pi}{5}\sin\frac{2\pi}{5}\sin\frac{3\pi}{5}\sin\frac{4\pi}{5}$$

Then find value of : $[\alpha]^2 + [8\beta] + [3\gamma] + [16\delta] =$ _____.
(where [.] represents GIF)

9

$$\delta = \sin\frac{\pi}{5}\cdot\sin\frac{2\pi}{5}\cdot\sin\frac{2\pi}{5}\cdot\sin\frac{\pi}{5}$$

$$= \left(\sin\frac{\pi}{5}\cdot\sin\frac{2\pi}{5}\right)^2 = \left(\sin 36^\circ \times \sin 72^\circ\right)^2 = \left(\sin 36^\circ \cos 18^\circ\right)^2 = \sin^2 36 \cdot \cos^2 18^\circ$$

$$= \left(1-\cos^2 36^\circ\right)\left(1-\sin^2 18^\circ\right) = \frac{5}{16}$$

PRODUCT OF COSINE WITH DOUBLE ANGLES

$$\cos\theta \cdot \cos 2\theta \cdot \cos 4\theta \cdot \cos 8\theta \cdot \ldots \cos(2^{n-1}\theta) = \frac{\sin(2^n\theta)}{2^n \sin\theta}$$

Total Terms = n

θ = smallest angle

angle → double

func^n → cos

PRODUCT OF COSINE WITH DOUBLE ANGLES

(i) $\cos\theta \cdot \cos 2\theta = \frac{\sin(4\theta)}{2^2 \sin\theta}$

(ii) $\cos\theta \cdot \cos 2\theta \cdot \cos 4\theta = \frac{\sin 8\theta}{2^3 \cdot \sin\theta}$

(iii) $\cos\theta \cdot \cos 2\theta \cdot \cos 4\theta \cdot \cos 8\theta = \frac{\sin(16\theta)}{2^4 \sin\theta}$

(iv) $\cos\theta \cdot \cos 2\theta \cdot \cos 4\theta \cdot \cos 8\theta \cdot \ldots\ldots \cdot \cos(64\theta) = \frac{\sin(128\theta)}{2^7 \sin\theta}$

2^1 2^2 2^3 2^6

चिंगारी Series.

[IIT-JEE 1983]: $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{4\pi}{7} =$ _____

[JEE MAINS-2023]: $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{3\pi}{7} =$ _____.

PRODUCT OF COSINE WITH DOUBLE ANGLES

$$\cos\frac{\pi}{7}\cdot\cos\frac{2\pi}{7}\cdot\cos\frac{4\pi}{7} = \frac{-1\ \sin\frac{8\pi}{7}}{2^3\cdot\sin\frac{\pi}{7}} = \frac{-1}{8}$$

$$\sin\left(\pi+\frac{\pi}{7}\right) = -\sin\frac{\pi}{7}$$

[IIT-JEE 1983]: $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{4\pi}{7} = \underline{\quad -\frac{1}{8} \quad}$.

$$\cos\frac{\pi}{7}\cdot\cos\frac{2\pi}{7}\cdot\left(-\cos\frac{4\pi}{7}\right)$$

$$-\left(-\frac{1}{8}\right)$$

[JEE MAINS-2023]: $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{3\pi}{7} = \underline{\quad +\frac{1}{8} \quad}$.

$$\cos\left(\pi-\frac{4\pi}{7}\right) = -\cos\frac{4\pi}{7}$$

$$\cos\theta \cdot \cos 2\theta = \frac{2\sin\theta\cos\theta \cdot \cos 2\theta}{2\sin\theta} = \frac{2\sin 2\theta \cdot \cos 2\theta}{2\times 2\sin\theta} = \frac{\sin(4\theta)}{2^2\cdot\sin\theta}.$$

$$\cos\theta \cdot \cos 2\theta \cos 4\theta = \frac{2\sin\theta\cos\theta}{2\sin\theta}\cos 2\theta\cos 4\theta = \frac{2\sin 2\theta\cdot\cos 2\theta}{2\times 2\sin\theta}\cos 4\theta = \frac{2\sin 4\theta\cdot\cos 4\theta}{2\times 2^2\sin\theta}.$$

$$= \frac{\sin 8\theta}{2^3\sin\theta}.$$

Question [JEE Mains-2019]

"9 Terms"

The value of $\cos\frac{\pi}{2^2}\cdot\cos\frac{\pi}{2^3}\cdot\ldots\ldots\cdot\cos\frac{\pi}{2^{10}}\sin\frac{\pi}{2^{10}}$ is :

A $\frac{1}{1024}$

B $\frac{1}{512}$

C $\frac{1}{2}$

D $\frac{1}{256}$

$$\frac{\sin\left(\frac{2\pi}{2^{10}}\right)}{2^9\sin\frac{\pi}{2^2}}$$

✗ Wrong

$\theta \rightarrow$ smallest angle

$$\cos\frac{\pi}{2^{10}}\cdot\cos\frac{\pi}{2^9}\cdot\cos\frac{\pi}{2^8}\cdot\ldots\ldots\cdot\cos\frac{\pi}{2^3}\cdot\cos\frac{\pi}{2^2}$$

$$\frac{1}{2^9}=\frac{\sin\left(2\times\frac{\pi}{2^2}\right)}{2^9\sin\left(\frac{\pi}{2^{10}}\right)}\times\sin\frac{\pi}{2^{10}}$$

Question [JEE Mains-2023]

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$96\cos\frac{\pi}{33}\cos\frac{2\pi}{33}\cos\frac{4\pi}{33}\cos\frac{8\pi}{33}\cos\frac{16\pi}{33}$ is equal to

A 4

B 2

C 1

D 3

$\# \ \frac{\pi}{33} = \theta$.

$96\cos\theta\cdot\cos 2\theta\cdot\cos 4\theta\cdot\cos 8\theta\cdot\cos 16\theta$

$\sin\left(\pi - \frac{\pi}{33}\right) = \sin\frac{\pi}{33}$.

$\frac{32\pi}{33}$

$96 \times \frac{\sin\left(32\frac{\pi}{33}\right)}{2^5\sin\frac{\pi}{33}} = \frac{96}{32} = 3$

Question [JEE Mains-2022]

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$2\sin\left(\frac{\pi}{22}\right)\sin\left(\frac{3\pi}{22}\right)\sin\left(\frac{5\pi}{22}\right)\sin\left(\frac{7\pi}{22}\right)\sin\left(\frac{9\pi}{22}\right)$ is equal to



A $\frac{3}{16}$

B $\frac{1}{16}$

C $\frac{1}{32}$

D $\frac{9}{32}$

CONDITIONAL IDENTITIES

Given cond^n: $(A+B+C=\pi)$

Involving SINE & COSINE:

1. $\sin 2A + \sin 2B + \sin 2C = 4\sin A \sin B \sin C$

2. $\cos 2A + \cos 2B + \cos 2C = -1 - 4\cos A \cos B \cos C$

3. $\sin A + \sin B + \sin C = 4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$

4. $\cos A + \cos B + \cos C = 1 + 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

RAR

CONDITIONAL I.D.

A, B, C are angles of Δ.

$$A + B + C = \pi$$

$$A + B = \pi - C$$

$$C = \pi - (A+B)$$

$$\cos C = -\cos(A+B)$$

$$\sin(A+B) = \sin(\pi - C)$$

$$= \sin C.$$

• Show

$$\#\ \sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C.$$

$$2\sin(A+B)\cos(A-B) + 2\sin C \cdot \cos C$$

$$2\sin C\left\{\cos(A-B) - \cos(A+B)\right\}$$

$$2\sin C \cdot 2\sin A \sin B.$$

HHPP.

CONDITIONAL I.D.

A, B, C are angles of Δ.

$$A + B + C = \pi$$

$$\cos\left(\frac{A+B}{2}\right) = \cos\left(\frac{\pi - C}{2}\right)$$

$$= \cos\left(\frac{\pi}{2} - \frac{C}{2}\right)$$

$$= \sin\frac{C}{2}$$

• Show

$$\cos A + \cos B + \cos C = 1 + 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$$

$$2\cos\frac{A+B}{2}\cos\frac{A-B}{2} + 1 - 2\sin^2\frac{C}{2}$$

$$1 + 2\sin\frac{C}{2}\cos\left(\frac{A-B}{2}\right) - 2\sin^2\frac{C}{2}$$

C D

$$1 + 2\sin\frac{C}{2}\left\{\cos\left(\frac{A-B}{2}\right) - \cos\left(\frac{A+B}{2}\right)\right\}$$

$$1 + 2\sin\frac{C}{2}\cdot 2\sin\left(\frac{A}{2}\right)\sin\left(\frac{B}{2}\right)$$

CONDITIONAL IDENTITIES

Given: $A+B+C=\pi$

Involving TANGENT & COTANGENT:

$(\tan A + \tan B + \tan C = \tan A \tan B \tan C)$

$(\cot A \cot B + \cot B \cot C + \cot C \cot A = 1)$

Same

$\tan 2A + \tan 2B + \tan 2C = \tan 2A \tan 2B \tan 2C$

$$\left(\tan\frac{A}{2}\tan\frac{B}{2}+\tan\frac{B}{2}\tan\frac{C}{2}+\tan\frac{C}{2}\tan\frac{A}{2}\right)=1$$

CONDITIONAL I.D.

A, B, C are angles of Δ.

$$A + B + C = \pi$$

multiply by (n) → Natural No.

$$nA + nB + nC = n\pi$$

$$nA + nB = n\pi - nC$$

$$\tan(nA + nB) = \tan(n\pi - nC)$$

$$\frac{\tan(nA) + \tan(nB)}{1 - \tan(nA)\tan(nB)} = -\tan nC$$

$$\tan(nA) + \tan(nB) = -\tan(nC) + \tan(nA)\tan(nB)\tan(nC)$$

$$\tan(nA) + \tan(nB) + \tan(nC) = \tan(nA)\tan(nB)\tan(nC)$$

$n\pi - nC = n\pi - \theta$

$\tan(n\pi - \theta) = -\tan\theta = -\tan(nC)$

$\tan(1\pi - \theta) = -\tan\theta$

$\tan(2\pi - \theta) = -\tan\theta$

$\tan(3\pi - \theta) = -\tan\theta$

$\tan(4\pi - \theta) = -\tan\theta$

⋮

CONDITIONAL I.D.

A, B, C are angles of Δ.

$$A + B + C = \pi$$

$$\tan(nA) + \tan(nB) + \tan(nC) = \tan(nA)\tan(nB)\tan(nC), \quad n \in N.$$

$n = 1$ $\quad \tan(A) + \tan(B) + \tan(C) = \tan(A)\tan(B)\tan(C)$

$n = 2$ $\quad \tan(2A) + \tan(2B) + \tan(2C) = \tan(2A)\tan(2B)\tan(2C)$

$n = 3$ $\quad \tan(3A) + \tan(3B) + \tan(3C) = \tan(3A)\tan(3B)\tan(3C)$

⋮

CONDITIONAL I.D.

A, B, C are angles of Δ.

$$A + B + C = \pi$$

multiply $\frac{1}{2}$

$$\frac{A}{2} + \frac{B}{2} = \frac{\pi}{2} - \frac{C}{2}$$

$$\tan\left(\frac{A}{2} + \frac{B}{2}\right) = \tan\left(\frac{\pi}{2} - \frac{C}{2}\right)$$

$$\frac{\tan\frac{A}{2} + \tan\frac{B}{2}}{1 - \tan\frac{A}{2}\tan\frac{B}{2}} = \frac{1}{\tan\frac{C}{2}}$$

आओ याद करें

sine & cos

⊕ ⟶ multip. ✓

sin ⟶ 4 ⏦

cos ⟶ 1 4 ⏦

2A ⟶ A (funcⁿ. change ✗)

A ⟶ $\frac{A}{2}$ (funcⁿ. change ✓)

$\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$.

$\sin A + \sin B + \sin C = 4 \cos\frac{A}{2} \cos\frac{B}{2} \cos\frac{C}{2}$

$\cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C$

$\cos A + \cos B + \cos C = 1 + 4 \sin\frac{A}{2} \sin\frac{B}{2} \sin\frac{C}{2}$

Question Practice

Question

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Show that : If $A + B + C = \pi$ then $\sin^2 A + \sin^2 B + \sin^2 C = 2(1 + \cos A \cos B \cos C)$

$\sin^2\theta = \frac{1-\cos 2\theta}{2}$

$$\text{LHS} = \frac{1}{2} - \frac{\cos 2A}{2} + \frac{1}{2} - \frac{\cos 2B}{2} + \frac{1}{2} - \frac{\cos 2C}{2}$$

$$= \frac{3}{2} - \frac{1}{2}\left(\cos 2A + \cos 2B + \cos 2C\right)$$

$$= \frac{3}{2} - \frac{1}{2}\left(-1 - 4\cos A \cos B \cos C\right)$$

$$= 2 + 2\cos A \cos B \cos C$$

HPP.

Challenger Question

If $\cos^{-1}a + \cos^{-1}b + \cos^{-1}c = \pi$, then the value of $\left(a\sqrt{1-a^2} + b\sqrt{1-b^2} + c\sqrt{1-c^2}\right)$ will be

A. $2\left(\sqrt{1-a^2}\sqrt{1-b^2}\sqrt{1-c^2}\right)$

B. $\sqrt{1-a^2}\sqrt{1-b^2}\sqrt{1-c^2}$

C. abc

D. $2abc$

$a\sqrt{1-a^2} = \cos A \cdot \sqrt{1-\cos^2 A}$

$= \cos A \cdot |\sin A|$

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$= \cos A \cdot \sin A$

$= \frac{\sin 2A}{2}$

Say $\begin{cases} \cos^{-1}a = A \Rightarrow a = \cos A \\ \cos^{-1}b = B \Rightarrow b = \cos B \\ \cos^{-1}c = C \Rightarrow c = \cos C \end{cases}$

$\sin A = \sqrt{1-\cos^2 A}$

$= \sqrt{1-a^2}$

Given: $A + B + C = \pi$ ✓

Asked $= \frac{\sin 2A}{2} + \frac{\sin 2B}{2} + \frac{\sin 2C}{2} = \frac{1}{2}(2\sin A \sin B \sin C)$

$= 2\sqrt{1-a^2}\sqrt{1-b^2}\sqrt{1-c^2}$

If $xy + yz + zx = 1$ where $x, y, z \in R^+$, then prove that

$$\frac{2x}{1+x^2} + \frac{2y}{1+y^2} + \frac{2z}{1+z^2} = \frac{2 \times 2}{\sqrt{(1+x^2)(1+y^2)(1+z^2)}}$$

Put

$x = \tan\frac{A}{2}$

$y = \tan\frac{B}{2}$

$z = \tan\frac{C}{2}$

Given. $A+B+C = \pi$.

To prove.

$\sin A + \sin B + \sin C = 4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$ ✓

if $A+B+C = \pi$

$\tan\frac{A}{2}\tan\frac{B}{2} + \tan\frac{B}{2}\tan\frac{C}{2} + \tan\frac{C}{2}\tan\frac{A}{2} = 1$

$x = \tan\frac{A}{2}$

$$\frac{2x}{1+x^2} = \frac{2\tan\frac{A}{2}}{1+\tan^2\frac{A}{2}} = \sin 2\left(\frac{A}{2}\right) = \sin A.$$

$$\sqrt{1+x^2} = \sqrt{1+\tan^2\frac{A}{2}} = \sqrt{\sec^2\frac{A}{2}} = \left|\sec\frac{A}{2}\right| = \sec\frac{A}{2}$$

Question

If $A + B + C = \pi$, then the value of expression $\frac{\cos A}{\sin B \sin C} + \frac{\cos B}{\sin C \sin A} + \frac{\cos C}{\sin A \sin B}$ is

Ans.

Question

If $\cos^2 A + \cos^2 B + \cos^2 C = 1$, then $\triangle ABC$ is

A Equilateral

B Isosceles

C Right angled

D None of these

Question

In $\triangle ABC$, if $\cot A + \cot B + \cot C = 0$ then find the value of $\cos A \cos B \cos C$.

Hint

Question

Let A, B, C be the three angles such that $A + B + C = \pi$. If $\tan A \cdot \tan B = 2$, then find the value of $\frac{\cos A \cos B}{\cos C}$.

Ans.

DO IT BY YOURSELF (DIBY)

DIBY-23

Prove that

10. $\cos(A+B)+\sin(A-B)=2\sin(45^\circ+A)\cos(45^\circ+B)$

$\sin(45^\circ+A+45^\circ+B)+\sin(45^\circ+A-(45^\circ+B))$

$90^\circ+(A+B)$

$\cos(A+B)+\sin(A-B)$

DIBY-23

Prove that

11. $\dfrac{\cos 3A-\cos A}{\sin 3A-\sin A}+\dfrac{\cos 2A-\cos 4A}{\sin 4A-\sin 2A}=\dfrac{\sin A}{\cos 2A\cos 3A}$

$$\frac{-2\sin\left(\frac{3A+A}{2}\right)\sin\left(\frac{3A-A}{2}\right)}{2\cos\left(\frac{3A+A}{2}\right)\sin\left(\frac{3A-A}{2}\right)}+\frac{2\sin\left(\frac{2A+4A}{2}\right)\sin\left(\frac{2A-4A}{2}\right)}{2\cos\left(\frac{4A+2A}{2}\right)\sin\left(\frac{4A-2A}{2}\right)}$$

(+ve)

$$-\frac{\sin 2A}{\cos 2A}+\frac{\sin 3A}{\cos 3A}=\frac{-\sin 2A\cos 3A+\sin 3A\cos 2A}{\cos 2A\cos 3A}=\sin(3A-2A)$$

DIBY-24

Find the value of expression $\frac{\sin8\theta\cos\theta-\sin6\theta\cos3\theta}{\cos2\theta\cos\theta-\sin3\theta\sin4\theta}$, given $\theta = 7.5°$.

(A) $2+\sqrt{3}$ (B) $2-\sqrt{3}$ (C) 0 (D) 1

[Ans. : (B)]

$$N^r = \frac{1}{2}\left(2\sin\overset{A}{8\theta}\cos\overset{B}{\theta} - 2\sin\overset{A}{6\theta}\cos\overset{B}{3\theta}\right)$$

$$= \frac{1}{2}\left(\sin9\theta + \sin7\theta - (\sin9\theta + \sin3\theta)\right)$$

$$= \frac{1}{2}\left(\sin\overset{C}{7\theta} - \sin\overset{D}{3\theta}\right)$$

$$= \frac{1}{2}\left(2\sin5\theta\cos2\theta\right)$$

$$D^r = \frac{1}{2}\left(2\cos\overset{A}{2\theta}\cos\overset{B}{\theta} - 2\sin\overset{A}{3\theta}\sin\overset{B}{4\theta}\right)$$

$$= \frac{1}{2}\left(\cos\theta + \cos3\theta - (\cos\theta - \cos7\theta)\right)$$

$$= \frac{1}{2}\left(\cos\overset{C}{3\theta} + \cos\overset{D}{7\theta}\right)$$

$$= \frac{1}{2}\left(2\cos5\theta\cos2\theta\right)$$

$$\frac{N^r}{D^r} = \tan(5\theta)$$

$$= \tan(5\times7.5^\circ) = \tan(37.5^\circ)$$

DIBY-29

If $x = \frac{\pi}{48}$, find the value of $\frac{\overset{C}{\sin 7x} + \overset{D}{\sin 5x} + \overset{C}{\sin 9x} + \overset{D}{\sin 3x}}{\underset{C}{\cos 7x} + \underset{D}{\cos 5x} + \underset{C}{\cos 9x} + \underset{D}{\cos 3x}}$.

[Ans. : $\sqrt{2} - 1$]

$$\frac{2\sin 6x\cos x + 2\sin 6x\cos\frac{3x}{2}}{2\cos 6x\cos x + 2\cos 6x\cos\frac{3x}{2}} = \frac{2\sin 6x}{2\cos 6x}\frac{\left(\cos x + \cos\frac{3x}{2}\right)}{\left(\cos x + \cos\frac{3x}{2}\right)}$$

$$= \tan(6x) \quad \left(x = \frac{\pi}{48}\right)$$

$$= \tan\left(6 \times \frac{\pi}{48}\right)$$

$$= \tan\frac{\pi}{8}$$

DIBY-30

If $3\sin\alpha = 5\sin\beta \neq 0$, then the value of $\dfrac{\tan\left(\frac{\alpha+\beta}{2}\right)}{\tan\left(\frac{\alpha-\beta}{2}\right)}$ is:

[Ans. : 4]

$$\frac{\sin\alpha}{\sin\beta} = \frac{5}{3}$$

apply componendo & dividendo

$$\frac{\sin\alpha + \sin\beta}{\sin\alpha - \sin\beta} = \frac{5+3}{5-3}$$

$$\frac{2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)}{2\cos\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)} = \frac{8}{2} = 4$$

DIBY-31

The value of $\cos 24^\circ - \cos 12^\circ + \cos 48^\circ - \cos 84^\circ =$

[Ans. : 1/2]

$$(\cos 24^\circ - \cos 84^\circ) + (\cos 48^\circ - \cos 12^\circ)$$

$$-2\sin\left(\frac{24^\circ+84^\circ}{2}\right)\sin\left(\frac{24^\circ-84^\circ}{2}\right) + \left(-2\sin\left(\frac{48^\circ+12^\circ}{2}\right)\sin\left(\frac{48^\circ-12^\circ}{2}\right)\right)$$

54°, -30°, 30°, 18°

$$2\sin 54^\circ \times \sin 30^\circ - 2 \times \frac{1}{2} \times \sin 18^\circ$$

$$\cos 36^\circ - \sin 18^\circ = \frac{\sqrt{5}+1}{4} - \left(\frac{\sqrt{5}-1}{4}\right) = \frac{2}{4} = \frac{1}{2}$$

DIBY-32 error!

If $\cos\alpha + \cos\beta = \frac{1}{2}$ and $\sin\alpha + \sin\beta = \frac{1}{3}$

Column - I	Column - II
(A) $\cos\left(\frac{\alpha+\beta}{2}\right)$	(P) $\pm\frac{\sqrt{13}}{12}$
(B) $\cos\left(\frac{\alpha-\beta}{2}\right)$	(Q) $\frac{2}{3}$
(C) $\tan\left(\frac{\alpha+\beta}{2}\right)$	(R) $\pm\frac{3}{\sqrt{13}}$
(D) $\tan\left(\frac{\alpha-\beta}{2}\right)$	(S) $\pm\sqrt{\frac{131}{13}}$

[Ans. : A-R,B-P,C-Q,D-S]

DIBY-33

Prove that $\frac{\sin\theta+\sin 2\theta}{1+\cos\theta+\cos 2\theta}=\tan\theta$

DIBY-34

Prove that $\frac{1-\cos A+\cos B-\cos(A+B)}{1+\cos A-\cos B-\cos(A+B)}=\tan\frac{A}{2}\cot\frac{B}{2}$

DIBY-35

Value of : $\prod_{k=1}^{17}\sin\frac{k\pi}{18}=\frac{a}{2^{16}}$ then $a=$ _____.

[Ans. : 9]

DIBY-36

If $\frac{5\pi}{2}<x<3\pi$, then the value of the expression $\frac{\sqrt{1-\sin x}+\sqrt{1+\sin x}}{\sqrt{1-\sin x}-\sqrt{1+\sin x}}$ is

(A) $-\cot\frac{x}{2}$ (B) $\cot\frac{x}{2}$ (C) $\tan\frac{x}{2}$ (D) $-\tan\frac{x}{2}$

[Ans. : (D)]

DIBY-34

Prove that $\dfrac{1-\cos A+\cos B-\cos(A+B)}{1+\cos A-\cos B-\cos(A+B)} = \tan\dfrac{A}{2}\cot\dfrac{B}{2}$

$$\text{N}^r = \underbrace{1-\cos A}_{\downarrow\downarrow} + \overbrace{\cos B}^{C} - \overbrace{\cos(A+B)}^{D}$$

$$2\sin^2\frac{A}{2} + 2\sin\left(\frac{B+A+B}{2}\right)\sin\left(\frac{B+A-B}{2}\right)$$

$$\underbrace{2\sin\frac{A}{2}}\left\{\underbrace{\sin\frac{A}{2}}_{C} + \underbrace{\sin\left(\frac{A}{2}+B\right)}_{D.}\right\}$$

$$\text{D}^r = \underbrace{1+\overbrace{\cos A}^{2\cos^2\frac{A}{2}-1}}_{\downarrow} - \left(\overbrace{\cos B}^{C} + \overbrace{\cos(A+B)}^{D}\right)$$

$$= \underbrace{2\cos^2\frac{A}{2}} - 2\cos\left(\frac{B+A+B}{2}\right)\cos\left(\frac{B-A-B}{2}\right)$$

$$= \underbrace{2\cos\frac{A}{2}}\left\{\overbrace{\cos\frac{A}{2}}^{C} - \overbrace{\cos\left(\frac{A}{2}+B\right)}^{D.}\right\}$$

DIBY-37

If $A = 340^\circ$ then $2\sin\frac{A}{2}$ is identical to

(A) $\sqrt{1+\sin A}+\sqrt{1-\sin A}$

(B) $-\sqrt{1+\sin A}-\sqrt{1-\sin A}$

(C) $\sqrt{1+\sin A}-\sqrt{1-\sin A}$

(D) $-\sqrt{1+\sin A}+\sqrt{1-\sin A}$

[Ans. : (D)]

DIBY-38

Evaluate: $\cos^6\frac{\pi}{16}+\cos^6\frac{3\pi}{16}+\cos^6\frac{5\pi}{16}+\cos^6\frac{7\pi}{16}$

(A) 5/8 (B) 5/4 (C) 7/16 (D) 5/32

[Ans. : (B)]

DIBY-39

If $\frac{x}{\cos\theta}=\frac{y}{\cos\left(\theta+\frac{2\pi}{3}\right)}=\frac{z}{\cos\left(\theta-\frac{2\pi}{3}\right)}$, then $x+y+z$ is equal to:

(A) –1 (B) 1 (C) 0 (D) 2

[Ans. : (C)]

DIBY-40

Find the exact value of $\tan^2\frac{\pi}{16}+\tan^2\frac{3\pi}{16}+\tan^2\frac{5\pi}{16}+\tan^2\frac{7\pi}{16}$

[Ans. : 28]

DIBY-41

Prove that $\cos36^\circ\cos72^\circ\cos108^\circ\cos144^\circ = 1/16$.

DIBY-42

If $A+B+C=\pi$, prove that $\sin^2\frac{A}{2}+\sin^2\frac{B}{2}-\sin^2\frac{C}{2}=1-2\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$.

DIBY-43

If $A+B+C=180^\circ$, prove that $\cos^2A+\cos^2B+\cos^2C=1-2\cos A\cos B\cos C$.



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians